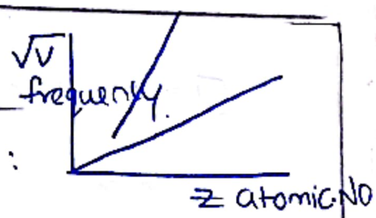
LESSON-2

PERIODIC CLASSIFICATION



Henry Moseley :

IUPAC

0	nil	n
1	un	u
2	bi	b
3	tri	t
4	quad	q
5	Pent	p
6	hex	h
7	Sept	s
8	Oct	o
9	enn	e

Pauling Scale:

$$\Delta = X_A - X_B = 0.208\sqrt{E_{A-B} - \sqrt{E_{AA} \times E_{BB}}}$$

Mulliken's Scale :

$$X_M = \frac{IE + EA}{2}$$

$$x_P = 1.35\,(x_M)^{1/2} - 1.37.$$

$\boxed{\text{Mulliken} = 2.8 \times \text{Pauling's Value}}$

$$\text{Electron Affinity} \propto \frac{1}{\text{At. Size}}$$

$$\propto \text{Nuclear Charge}$$

$$\propto \frac{1}{\text{screening effect}}$$

$$r_{covalent} < r_{crystal} < r_{Vanderwalls}$$

- Johann Dobereiner : 1829 (Triads)
- Newlands : 1865 (Law of octaves)
- Dechancourtois : 1862 (Increasing At wt in Cylinderical form)
- Lother Meyer : Physical Properties in Repeated Pattern
- Mendeleev (101 element) : Similarities in Empirical formulas / Properties. → Eka Aluminium / Eka Silicon (Left GAP)
- ~~1108~~ Henry Mosely :
  - 94 Nature Occurance.
  - x-ray Spectra ..
  - Graph

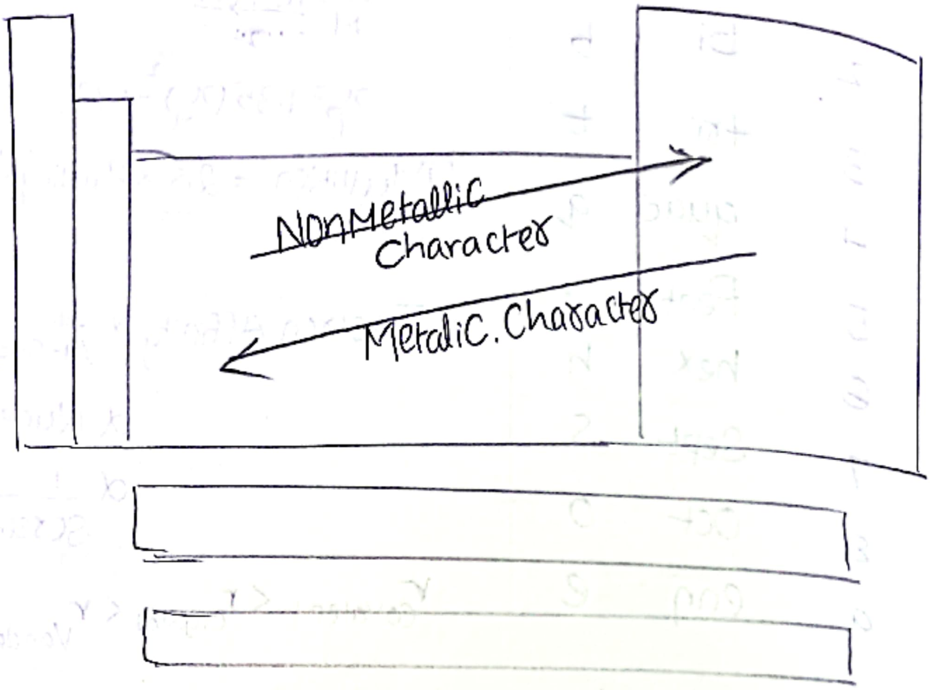
B/w

in 1800	–	31 elements
In 1865	–	63 elements
In Present	–	114 elements ...

$\boxed{\sqrt{\text{frequency (X ray)}}\ \text{vs At.wt}}$

Electronegativity → Group : (Left to Right) Increase
→ Period (Top bottom) Decrease.

Electropositivity → Group : (Left Right) Decrease
→ Period : (Top bottom) Increase.



- Effictive Nuclear Charge donoted by Z_{eff}
- Oxidation $\propto$ Acidic.

Overlapping :

S–S
P–P
d_{xy}–d_{xy}

i. a Sigma bond

ii. π Bonds :

iii. δ Bonds. Delta Bond

Formal Charge :

$= V - U - \frac{S}{2}$

where

V → No. of Valence $\bar{e}$
U → No. of nonbondi
S → No. of Bonding or Shared $\bar{e}$

Exceptions in Periodic Table

		What we think	But Actual
1.	EA :	$F > Cl > Br > I$	$Cl > F > Br > I$
2.	EA :	$O > S > Se > Te > Po$ $S > O > Se > Te > Po$	$S > Se > Te > Po > O$
3.	BDE :	$F_2 > Cl_2 > Br_2 > I_2$	$Cl_2 > Br_2 > F_2 > I_2$
4.	D.M :	$CH_3F > CH_3Cl > CH_3Br > CH_3I$	$CH_3Cl > CH_3F > CH_3Br > CH_3I$
5.	BA	—	$OCl_2 > OH_2 > OF_2$
6.	B.P	$HF < HCl < HBr < HI$	$HCl < HBr < HI < HF$
7.	B.P	$NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$	$PH_3 < AsH_3 < NH_3 < SbH_3 < BiH_3$
8.	B.P	$H_2O < H_2S < H_2Se < H_2Te$	$H_2S < H_2Se < H_2Te < H_2O$
9.	IE (2nd Period)	$Li < Be < B < C < N < O < F < Ne$	$Li < B < Be < C < O < N < F < Ne.$
10.	IE (3rd Period)	—	$Na < Al < Mg < Si < S < P < Cl < Ar$
11.	AR	$Ti < Zr < Hf$	$Ti < Zr \approx Hf$

NOTE : Applicable 4, 5, 6, 7, 8, 9, 10, 11, 12

"NOTE":

Exception : $Sc < Y < La$ $Sc < Y < La$.

		What we think	But Actual
13.	AR	$B < Al < Ga < In < Tl$	$B < Ga < Al < In < Tl$

14. Sodium Nitroprusside : $Na_2[CN_5 NO]$ $NO \rightarrow +1$, $Fe \rightarrow +2$

15. Brown Ring Complex : $[Fe(H_2O)_5NO]SO_4$ $NO \rightarrow +1$ $Fe \rightarrow +3$

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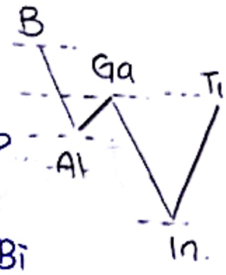
16. IE : — $\boxed{B > Tl > Ga > Al > In}$



17. Group 14 IE $C > Si > Ge > Sn > Pb$ $C > Si > Ge > Sn < Pb$

18. Group 15 No Exception $N > P > As > Sb > Bi$

19. IE : 3d series $Sc < Ti < V < Cr < Mn < Fe < Co < Ni < Cu < Zn$ $Sc < Ti > V < Cr < Mn < Fe > Co > Ni < Cu < Zn \rightarrow V, Co, Ni$

$[PtCl_4]^{2-}[PdCl_4]^{2-} \rightarrow$ C.NO : 4 SFL / WFL } dsp^2 : diamagnetic Nonpolar

$[Co(H_2O)_6]^{3+}[Co(ox)_3]^{3-}$: d^2sp^3 Oxygen donar in +3 SFL.

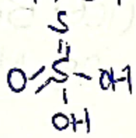
SF_6 : Kinetically inert $\rightarrow$ Can't Undergo hydrolysis

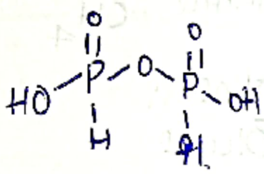
$BCl_3 + Cl^-$: No Reactions
$BI_3 + I^-$ " "
} Doesn't Exisist $BF_3 + F^- \rightarrow BF_4^-$ Acts as Lewis Acid.

24. BDE : $C-C > Si-Si$ $O-O < S-S$ ~~C-C > Si-Si~~ ~~O-O < S-S~~
$N-N < P-P$ $F-F < Cl-Cl$ ~~N≡N < P-P~~ ~~F-F > O-O~~

25 BDE : $C-C > N-N > F-F > O-O$

26. DM : $C-Cl > C-F$.

27. $H_2S_2O_3$  ✓ :

28. $H_4P_2O_5 \longrightarrow$ Dibasic Acid : 

29. IH_7 PH_5 SH_4 XeH_2 XeH_4 : doesn't Exisist IF_7 PF_5 SF_4 SF_6 XeF_2 XeF_4 Exisist.

30. Alkali metals + N_2 No reaction But Li air $\longrightarrow Li_2O + Li_3N$.

31. Acidic Strength : $HClO_4 > HClO_3 > HClO_2 > HCl$ ON↑ se Acidic Nature↑se.

- $H_3PO_4 > H_3PO_3 > H_3PO_2$ ✗ $H_3PO_2 > H_3PO_3 > H_3PO_4$. Basicity of Acid.

- $PCl_{5(s)} \longrightarrow PCl_4^+ + PCl_6^-$ Sp^3 Sp^3d^2 $PCl_{5(g)} \longrightarrow Sp^3d$ only

Group I.

Density: Cs > Rb > Na > K > Li

Group II

IE : Be > Mg > Ca > Sr > Ra > Ba

MP : Be > Ca > Sr > Ba > Ra > Mg.

B.P : Be > Ba > Sr > Ca > Mg ...

f

F Block:

57 58 59 60 61 62 63 9 10 11 12 13 14 14

La Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu

Ac Th Pa U Np Pu Am Cm Bk Cf Es Fm Md No Lr

2 3 4 6 7 7 9 10 11 12 13 14 14

$f^1 f^5 f^8$

Sc_{21}

Y_{39}

La_{57}

Oleum — $H_2S_2O_7$.

(60gr) Urea — NH_2CONH_2

(180gr) Glucose – $C_6H_{12}O_6$

(16gr) Methane – CH_4

(93gr) Aniline :

(92gr) Toluene :

(78) : Benzene & (123gr) Nitrobenzene

(112) Chlorobenzene.

(106) Benzaldehyde.

(128) Napthalene :